Characterization and automated classification of sentences in the biomedical literature: a case study for biocuration of gene expression and protein kinase activity

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Abstract

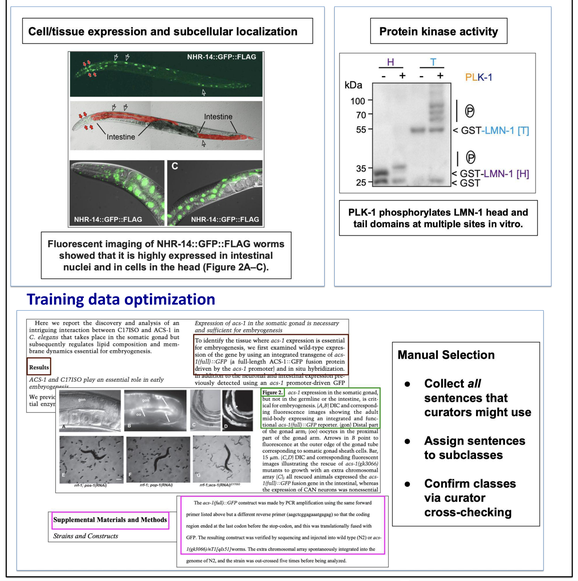
Biological knowledgebases are essential resources for biomedical researchers, providing ready access to gene function and genomic data. Professional, manual curation of knowledgebases, however, is labor-intensive and thus high-performing machine learning methods that improve biocuration efficiency are needed. Here we report on sentence-level classification to identify biocuration-relevant sentences in the full text of published references for two gene function data types: gene expression and protein kinase activity. We performed a detailed characterization of sentences from references in the WormBase bibliography and used this characterization to define three tasks for classifying sentences as either 1) fully curatable, 2) fully and partially curatable, or 3) all language-related. We evaluated various machine learning (ML) models applied to these tasks and found that GPT and BioBERT achieve the highest average performance, resulting in F1 performance scores ranging from 0.89 to 0.99 depending upon the task. Our findings demonstrate the feasibility of extracting biocuration-relevant sentences from full text. Integrating these models into professional biocuration workflows, such as those used by the Alliance of Genome Resources and the ACKnowledge community curation platform, might well facilitate efficient and accurate annotation of the biomedical literature.

ACKnowledge overview

ACKnowledge – Author Curation to Knowledgebases – combines artificial intelligence and community curation.

Aiding authors in fact extraction

Datatypes - Gene Expression and Protein Kinase Activity



Identification of sentence classes

1. Sentences that directly report an experimental result
   a. Fully curatable
      i. *A SNA-3::GFP translational fusion is expressed in the germline*
      ii. *AIR-2 phosphorylates TLK-1 at serine 634 in vitro*
   b. Partially curatable
      i. *The translational fusion was expressed in the germline*
      ii. *The fusion protein has functional kinase activity on a peptide substrate*
2. Summary statements
   a. Fully curatable
      i. *In summary, we show that SNA-3 is expressed in the germline*
      ii. *These results demonstrate that NSY-1 phosphorylates SEK-1*
   b. Partially curatable
      i. *Our work shows the localization of SNA-3*
      ii. *We have shown that the kinase domain is functional in vitro*
   c. Previously published
      i. *Smith et al have shown that sna-3 localizes in the germline*
      ii. *As reported previously, TLK-1 is an active serine/threonine protein kinase*
3. Other language related sentences
   a. Sentences that contain an hypothesis/rationale
      i. *We hypothesized that sna-3 was expressed…*
      ii. *We tested if SGK-1 was able to phosphorylate DAF-16*
   b. Sentences that describe experimental details or set up
      i. *The SNA-3::GFP translational fusion was constructed by inserting the sna-3 promoter…*
      ii. *The fusion proteins were immunopurified and used in kinase assays*
   c. Sentences that have related language but don't have curatable data
      i. *Psna-3 promoter was used for germline-specific expression*
      ii. *The active sites of protein kinases are shown in this alignment*

Gold Standard set
589 sentences from 115 papers for gene expression
494 sentences from 51 papers for protein kinase activity

Identification of tasks

1. Sentences that directly report an experimental result
   a. Fully curatable
      i. *A SNA-3::GFP translational fusion is ex*
      ii. *AIR-2 phosphorylates TLK-1 at serine*
   b. Partially curatable
      i. *The translational fusion was expresse*
      ii. *The fusion protein has functional kina*
2. Summary statements
   a. Fully curatable
      i. *In summary, we show that SNA-3 is ex*
      ii. *These results demonstrate that NSY-1 phosphorylates SEK-1*
   b. Partially curatable
      i. *Our work shows the localization of SNA-3*
      ii. *We have shown that the kinase domain is functional in vitro*
   c. Previously published
      i. *Smith et al have shown that sna-3 localizes in the germline*
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3. Other language related sentences
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   b. Sentences that describe experimental details or set up
      i. *The SNA-3::GFP translational fusion was constructed by inserting the sna-3 promoter…*
      ii. *The fusion proteins were immunopurified and used in kinase assays*
   c. Sentences that have related language but don't have curatable data
      i. *Psna-3 promoter was used for germline-specific expression*
      ii. *The active sites of protein kinases are shown in this alignment*

TASK 1 IDENTIFY FULLY CURATABLE SENTENCES
TASK 2 IDENTIFY PARTIALLY CURATABLE SENTENCES
TASK 3 IDENTIFY LANGUAGE RELATED SENTENCES

Identifying tasks utilizing relevant sentences

Training sets

589 sentences from 115 papers for gene expression
494 sentences from 51 papers for protein kinase activity

1000 randomly selected sentences

500 negative sentences

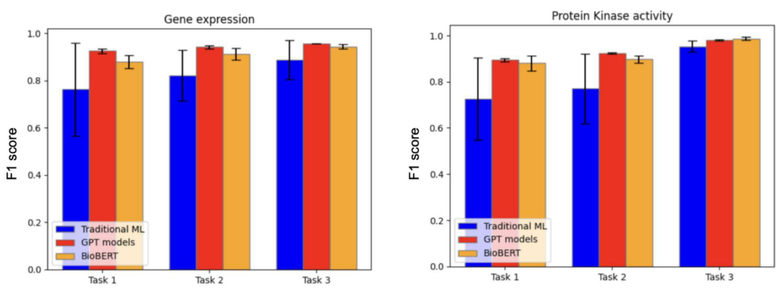
Classifiers training

Traditional Machine learning models
- Support Vector Machines
- Multi-Layer Perceptron
- Logistic regression
- Others

Determined the best model based on the F1 score

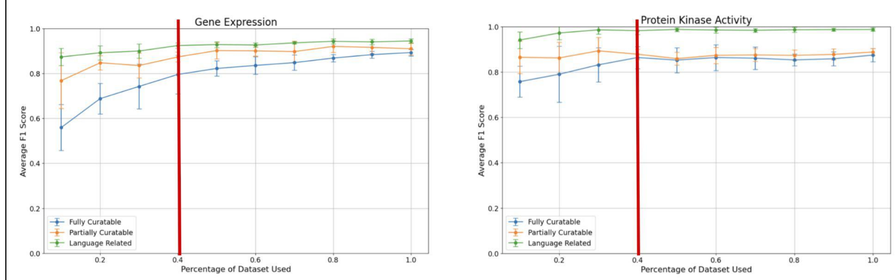
Large language Models
- BioBert
- GPT-4o

BioBert and GPT-4o outperformed traditional ML methods



GPT-4o and BioBERT showed strong performances, while traditional machine learning methods generally underperformed relative to these more advanced architectures.

Degradation analysis: How small can the training set be?



We performed a detailed analysis to assess the minimum amount of fine-tuning data required to achieve reliable performance while minimizing the time investment of curators. We see that even with 40% (200-250 sentences) of training data we achieve an F score ≥ 0.8.

Future plans

- Develop a UI to present sentences and facts to authors
  - Highlight sentences in the context of full text
  - Pre-populate curation forms
- Expand sentence classification to additional data types
- Generalize the tool to accommodate other organisms and literatures
- Funding: NIH grant #U24 HG002223